

# Specialization Is Not a Myth: Subject-Specific Expert Sets Recur During MoE Generation

Jeffrey Shorthill\*

## Abstract

In sparse Mixture-of-Experts models, a learned router selects a small subset of experts for each token from a fixed pool, during both prompt prefill and autoregressive generation. Prior work correctly finds that prompt-side routing can converge on one shared set of experts across domains. We show that generation gives a different answer.

We analyze five checkpoints across the Qwen3.5 and GLM-4.7-Flash families, prefill on all five and generation on the two Qwen3.5-35B-A3B checkpoints. In the HauhauCS fine-tune of Qwen3.5-35B-A3B-Instruct, we test whether the selected expert sets follow the local subject of an answer by packing multiple subjects into one generation, letting the model regroup them across prompts, controlling for position, and comparing them with the same questions asked independently. We repeat the subject test on the official instruction-tuned checkpoint from which the HauhauCS model was derived.

Position-resolved prefill routing primarily follows the wording of the question (Shorthill, 2026), while pooling across the prompt produces the shared usage pattern reported in prior work. Across packed and independently asked versions of identical questions, generation-based expert sets identify the subject at 95 to 96% in Qwen3.5-35B when the answer’s register is held constant (0.70 over all design pairs, 0.54 when the register changes), compared with 15 to 17% for prefill and 6% by chance. Generation-time expert sets remain stable within a subject, change sharply at subject boundaries, and recur for the same subject across positions, prompt design, and the official and fine-tuned checkpoints. On the official instruction-tuned checkpoint, generation-time sets pair prompts by subject in 11 of 12 cases where prefill pairs 4 of 12. For short mixed prompts, prefill carries the wording of the question, while generation

carries the subject of the answer.

## 1 Introduction

Expert specialization in sparse Mixture-of-Experts models (Jacobs et al., 1991; Shazeer et al., 2017; Fedus et al., 2022; Dai et al., 2024) is commonly inferred from which experts are selected for an input. That inference requires selective recurrence. A shared expert that appears across every subject may be important, but it is not specialized by subject. If the same expert set returns for the same subject after its wording, position, and surrounding context change, while different subjects recruit different sets, then the router is organizing computation by local subject. We use this subject-conditioned recurrence as the operational signature of specialization.

Prior work presents two views of expert organization. At the last prompt token of MMLU questions, the Standing Committee analysis finds that a small, fixed coalition carries most of the routing mass across nearly all domains (Wang et al., 2026b), echoing the Mixtral analysis, which observed no obvious topic-based assignment of experts on validation text (Jiang et al., 2024). Herbst et al. find that individual experts probe as fine-grained task specialists, such as experts associated with closing brackets or function definitions, rather than broad domain modules (Herbst et al., 2026). Ye et al. instead find that individual experts are polysemantic while their paths across layers become monosemantic (Ye et al., 2026). Do et al. identify domain-associated experts from question-token routing and show that up-weighting them can improve domain accuracy across ten models (Do et al., 2026). Routing statistics also track the input: task identity is recoverable from routing alone (Avinash, 2026), and routing during code generation on the checkpoint we use is strongly token- and layer-dependent (Hayashi et al., 2026). These findings place specialization at different units: individual experts, collaborative sets, or cross-layer paths.

The distinction between prompt prefill and text generation creates another divide; on the systems side the

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\*Independent researcher. Correspondence: jws299792@icloud.com. Version 1.2 (September 2026; version 1.1 August 2026, version 1.0 June 2026), released under CC BY 4.0. Concept DOI 10.5281/zenodo.20779604. Data, code, and a claim-by-claim source index accompany the paper; see Data and Code Availability.

two phases have distinct compute profiles and are increasingly served separately (Kwon et al., 2023; Zhong et al., 2024). Wang, Hayou, and Nalisnick find that semantically different prompts can converge on nearly identical expert usage during prefill and diverge during generation. Prompt-level routing therefore does not predict rollout-level routing, and complete trajectories are needed to characterize specialization (Wang et al., 2026a). Their result establishes the divergence, but leaves its organization unknown. The generated routes could follow the wording of the original question, move with token position, drift gradually through the answer, or separate according to the answer’s local subject. We ask which of these possibilities explains the rollout.

We test this question across five checkpoints from the Qwen3.5 and GLM-4.7-Flash families. The primary controlled experiment uses the HauhauCS fine-tune of Qwen3.5-35B-A3B-Instruct. We pack questions from multiple subjects into a single prompt, generate one answer containing each subject, use the model’s own regrouping of that answer to place subjects at different positions across prompts, and compare each generated segment with the same question asked independently. The same identification procedure is then applied to the prefill rows of those questions. This places the same semantic material on both sides of the prefill to generation boundary.

The design separates subject from four competing explanations. The model’s regrouping places the same subject at different token positions and different subjects at matched positions. Restricting the candidate set to the preceding, current, and following segments tests gradual drift. Comparing packed and independently asked versions changes prompt design and surrounding context. Holding the voice of the packed answer constant separates subject from register (the tone and verbiage of the text), while a second design allows register to vary. We repeat a smaller subject-matching experiment on the official instruction-tuned checkpoint to test whether the organization was already present before the HauhauCS fine-tune. The remaining checkpoints provide the prefill replication and model-family scope.

We reproduce the shared prefill pattern reported in prior work. When prefill routing is pooled over each question, one expert wins 18 of 20 subjects in Qwen3.5-35B. Position-resolved prefill routing primarily follows the wording and local form of the question: across packed and independently asked versions of the same questions, its expert sets identify the subject only 15 to 17% of the time. Generation gives a different answer. Expert-set similarity identifies the subject across those prompt designs at 95 to 96%, against 6% under shuffled labels.

The generation signal also survives the positional

and temporal controls. The same subject at different positions matches more strongly than different subjects at the same position in nine of ten comparisons. When the candidates are restricted to the previous, same, and next subjects, expert-set similarity still selects the correct subject 83% of the time, against 35% by chance. The sets therefore change at subject boundaries rather than sliding gradually through the answer. On the official instruction-tuned checkpoint, generation pairs prompts by subject in 11 of 12 cases, while prefill does so in 4 of 12. Recurring legal and medical experts also survive changes in dialect and the official to fine-tuned checkpoint boundary.

These results place the apparently conflicting findings of prior work into two routing regimes. For short mixed prompts, pooled prefill routing exposes a shared usage pattern, while position-resolved prefill routing remains closely tied to question form. During generation, local subjects recruit reproducible expert sets that remain stable within a subject, turn over at its boundary, and return when the subject reappears under another prompt design. This distinction is scoped rather than universal: on a long, coherent technical prompt, prefill and generation share much of the same expert set. The separation is strongest when a short question initiates a substantially longer answer organized around its subject.

We make four contributions. First, we replicate the shared prefill result on five checkpoints across two model families and two gating mechanisms, three of which no prior study had tested. Second, we define subject specialization through cross-design recurrence and provide a model-agnostic identification protocol with wording, position, drift, and shuffled-label controls. Third, we show that generation-time expert sets track the local subject of an answer at 95 to 96% identification accuracy, while prefill rows of the same questions reproduce the shared-routing result. Fourth, we show that this organization survives changes in position, prompt design, and the official instruction-tuned to fine-tuned checkpoint boundary. Register is a second organizer: subject survives when register is held constant, and a change of register moves the set as much as a change of subject does, lowering identification from 0.85 to 0.54. Expert specialization is visible when it is measured at the level of recurring generation-time sets.

## 2 Methods

**Models and captures.** We study five checkpoints. The primary architecture is Qwen3.5-35B-A3B, in the Qwen3 lineage (Yang et al., 2025), with 256 routed experts per layer, top-8 routing, and 40 MoE layers, in two versions: the official instruction-tuned checkpoint

(Q8\_0, ggml-org) and the HauhauCS uncensored fine-tune derived from it (Q8\_0). Each Qwen3.5 MoE layer also contains one always-active shared expert; the captured router logits cover only the 256 routed experts, and every expert set in this paper is over those. Qwen3.5-122B-A10B (HauhauCS, Q8\_K\_P) uses the same 256-expert, top-8 routing scheme with 48 MoE layers. Both Qwen3.5 models are hybrid stacks in which most layers use gated-DeltaNet token mixing in place of standard attention; every layer carries its own MoE router, and all routers were captured. GLM-4.7-Flash, official and HauhauCS, routes each token to 4 of 64 experts plus one shared expert by sigmoid gating with a per-layer correction bias.

For the Qwen checkpoints, router logits were captured at every MoE layer with a hook in llama.cpp under greedy decoding (temperature 0, top-k 1, fixed seed), storing routing only. Prompt and generated tokens were retained as separate blocks and were never combined. Generation blocks were cut at the first end-of-turn token when that token was available; the unrecoverable 122B exception is described under Provenance and scope. Generation was captured for both Qwen3.5-35B checkpoints. A 122B generation capture also exists but is treated as exploratory because its trimming failed. The GLM captures are prefill only and were taken from BF16 weights with a Transformers hook.

**Expert-set construction.** At each Qwen MoE layer, the router scores all 256 experts, and the scores are converted to probabilities by a softmax. The 8 highest were kept and rescaled to sum to 1, matching the weights the model uses to mix those experts. From these we construct the routed weight  $W$ , selection rate  $S$ , and conditional weight  $Q$ . For each token set,  $W$  was computed for every expert as the mean renormalized weight, with unselected experts contributing zero;  $S$  is the fraction of tokens on which the expert was selected,  $Q$  its mean weight when selected, and  $W = S \cdot Q$ . Layers were pooled by an unweighted mean over the per-layer  $W$  vectors, indexed by expert number. Because expert index  $e$  at one layer and expert index  $e$  at another layer denote different modules, these pooled objects are layer-pooled expert-index profiles rather than single experts persisting across depth. The primary claims therefore concern recurrence in layer-pooled routing organization. Per-layer identification is not available for the packed designs, whose surviving per-token tensors are layer-averaged; on the independent probe, where per-layer profiles survive, the generation-time dispersion is present at every layer, with about 14 distinct subject winners per layer across all 40 layers (prefill 1.6), slightly stronger in the middle third and weakest at the first and last layers (Appendix A). The layer-pooled

profile therefore summarizes a signal that is flat across depth rather than concentrated in a few layers. Prompt and generation blocks were never pooled, because the prompt block would dominate any combined set. The expert-index set of a token set is its top 8 indices by layer-pooled  $W$ . Sets are compared by the Jaccard index, whose exact expectation for two random sets of 8 from 256 is 0.017 (the ratio approximation  $k/(2E - k)$  gives 0.016). For GLM-4.7-Flash, the per-token rule is the model’s own: the top 4 experts by sigmoid score plus the per-layer correction bias, weighted by their plain sigmoid values rescaled to sum to 1; the remaining steps are unchanged with 64 experts and sets of 4.

**Prompt designs.** The independent probe asks about 20 subjects, with three questions per subject and one question per prompt; each answer was allowed up to 2,056 generated tokens. A subject’s probe profile is computed over its three answers. The packed designs place all 20 questions in one prompt and let the model answer them in one generation. Three packed prompts were used: A (explanations), B (biographies), and C (synthesis), each padded to 446 prompt tokens and allowed up to 2,048 generated tokens. The three prompts list the subjects in the same order; in C the model regrouped its answer under thematic headers, so its segments occur in a different order from A. Answer spans were assigned to subjects at the model’s own markdown headers, giving 12 segments in A, 15 in B, and 19 in C; the model skipped law in C and stopped early in A and B. Generation boundaries were taken from token IDs. No packed answer reached an end-of-turn token before the cap except C, which ended at token 1,812. A and C retain an expository register while subject changes. B changes the requested task and register while retaining the same subject labels, allowing register sensitivity to be measured across designs. For prefill spans, the Hugging Face tokenizer counts 391 to 392 tokens for the packed prompts where llama.cpp counted 446, so question spans were rescaled proportionally and may be off by one or two positions; generation spans are exact. The official instruction-tuned checkpoint design contains 12 prompts. Finance contributes two consequence-framed/neutral pairs (four prompts), while medical, physical, professional, and legal each contribute one pair (eight prompts); all answers were allowed 256 generated tokens. A 30-prompt set containing one long, coherent technical paragraph in three registers serves as a scoping analysis, not primary evidence.

**Cross-design identification.** For each subject segment we form its top-8 set by layer-pooled  $W$ . A query segment in design X is compared by Jaccard with every candidate segment in design Y, and the predicted

subject is the candidate with the highest similarity. Only subjects present in both designs enter either direction, and ties count as misses. Thus  $A \leftrightarrow P$  contributes  $12 + 12 = 24$  queries and  $C \leftrightarrow P$  contributes  $19 + 19 = 38$ . The pooled generation analysis contains 168 bidirectional queries after intersecting the subjects available in each design; Table 2 lists every ordered pair with its shared-subject count (A 12, B 15, C 19 and P 20 segments; law is absent from C, and A and B were capped by the generation budget). On prefill rows every design contains all 20 questions, so each pair contributes 40 queries and all six design pairs contribute 240. Identification accuracy is the fraction of queries whose predicted subject is correct; it is distinct from exact set recurrence, which we report separately as matched-subject Jaccard.

**Controls.** Shuffled-label null: subject labels were permuted across candidates 200 times (`NumPy default_rng(11)`) for generation and 100 times (seed 7) for prefill, and identification was recomputed; we report the mean, 95th percentile, and maximum. Position-matched null (Shorthill, 2026): A is compared with C, which the model answered in a different order, contrasting the same subject at different positions with different subjects at matched positions. Adjacent-segment null: candidates are restricted to the previous, same, and next subject in the target’s order; first and last segments have two candidates rather than three and were not excluded, so 21 of the 168 queries have two candidates and the exact chance level is 0.354. For queries whose target is the independent probe, which has no answer order, the query design’s order is used. Prefill control: the identical procedure is applied to the question-token rows of each design. Official-checkpoint replication: within-subject and between-subject Jaccard are compared across the crossed register conditions, followed by nearest-neighbour identification by subject among the other 11 prompts; because finance contributes four prompts, the chance level is 0.15. Set-size sensitivity: the prefill-versus-generation overlap was recomputed at top-8, top-15, and top-30; all three definitions gave the same qualitative conclusion.

**Causal bias test.** On the HauhauCS 35B, the four experts that lead philosophy in generation (114, 87, 170 and 68) received a common additive bias on their router logits before the top-8 selection, at every MoE layer, in nine runs with bias  $b \in \{-8, -5, -3, -2, 0, +2, +3, +5, +8\}$ , over the same 60 prompts with up to 1,024 generated tokens under greedy decoding. Negative bias at  $-2$  or below removes all four from every top-8 set; we report the per-subject winners, the per-subject top-8 overlap with the unbiased run, and the length and keyword content of the philosophy

answers. This is the only intervention in the paper.

**Statistical analysis.** Jaccard similarity is  $|A \cap B|/|A \cup B|$ . Identification accuracies are counts over the denominators above and are reported with two-sided 95% Wilson confidence intervals and against the shuffled-label distribution. Bidirectional queries share segments and are not independent; the intervals treat them as independent, and one-way counts are given in Table 2. No multiple-comparison correction is applied; the wording, position, drift, register, and shuffle controls were designed after the primary result and every one of them is reported. Within each captured segment, tokens contribute equally to  $W$ ; layers contribute equally to the layer-pooled profile; and each eligible segment contributes one identification query regardless of its length. For the independent probe,  $W$  was computed per answer and the subject profile is the unweighted mean over its three answers and 40 layers, so each answer carries equal weight regardless of its length.

**Provenance and scope.** Checkpoint files, hashes where recorded, llama.cpp builds, capture commands, prompt hashes, and per-run metadata are listed in the data repository’s `PROVENANCE.md`; the curated analysis repository is frozen at commit `f06cd3e`. The 35B fine-tune is SHA-256 `f3235db7...`, its 60-prompt domain probe used llama.cpp build 8493 (`1772701f`), the 122B capture binary has SHA-256 `c3e205b3...`, and the GLM register generation runs use commit `6658925`; the GLM domain prefill captures were taken with a Transformers hook and have no llama.cpp build. Per-token layer-averaged profiles survive for the packed prompts, and raw per-layer tensors survive for the controls and the GLM captures. The independent probe retains per-subject, per-layer summaries, and no 122B domain-probe tensors survive. Four exceptions determine the paper’s scope. First, the 122B generation trimmer was a no-op, leaving approximately 45% spill in its nominal generation block; that generation result is exploratory and is excluded from the primary replication claim. Second, the 122B generation-run prefill contains a last-layer artifact; a separate routing-only capture is clean and assigns 13 of 20 subjects to one expert. Third, GLM has no generation capture and supports only the prefill half of the analysis. Fourth, the official 35B checkpoint is instruction-tuned, so its comparison with HauhauCS tests persistence across that fine-tune, not emergence before instruction tuning. The primary packed-generation evidence therefore comes from the HauhauCS Qwen3.5-35B capture, with the smaller official instruction-tuned domain/register experiment serving as the generation-side checkpoint replication.

## 3 Results

### 3.1 Prefill expert sets reproduce across five checkpoints

Pooled over the prompt, one expert leads prefill routing for most subjects on every checkpoint we captured (Table 1). On the HauhauCS Qwen3.5-35B-A3B, expert 224 wins 18 of 20 subjects; the Herfindahl index of the winner distribution is 0.815 and the pairwise top-8 overlap between subjects averages 0.51. On Qwen3.5-122B-A10B, expert 233 wins 13 of 20 (routing-only capture); on this checkpoint the MoE layers with gated-DeltaNet token mixing and those with standard attention select disjoint top-8 expert vocabularies in every run, so the 48-layer pooled winner summarizes two routing systems. On the official instruction-tuned 35B checkpoint, expert 224 wins 6 of the 8 prompt types in the 50-pair run, and experts 95 and 224 together win 8 of the 12 prompts in the domain/register design. On GLM-4.7-Flash, with sigmoid gating, 64 experts and top-4 selection, expert 10 wins 12 of 20 subjects (15 prompts each), with five distinct winners and Herfindahl 0.41.

The GLM fine-tune serves as a control on what this default is made of. Its register battery varies the voice of the prompt with no subject axis, on both the official checkpoint and the HauhauCS fine-tune, whose router matrices are bit-identical. The pooled prefill top-4 set on the main register battery is {10, 20, 28, 51} on both checkpoints (overlap 1.0), and {16, 20, 28, 51} on the two secondary batteries, again identical across checkpoints. Expert 10, the winner on the subject battery, sits at rank 2 on the register battery. The same four or five experts carry prefill whether the prompts vary in subject or in voice, and whichever one wins on a given battery is a near-tie inside that set. The prefill default is a property of reading a prompt: subject-independent, fine-tune-independent, and present under both gating mechanisms.

This is the pattern the Standing Committee analysis reports on Qwen3-30B-A3B (cross-domain Jaccard 0.87, minimum 0.53) (Wang et al., 2026b). Our 0.51 is lower because it is computed over 20 subjects rather than nine pooled domains, over all prompt tokens rather than the last, and with routed weight rather than the full softmax; their minimum sits inside our range.

### 3.2 Generation disperses the same experts by subject

On the same 60 prompts, generation-time routing has 20 distinct subject winners on the 35B (Herfindahl 0.05), and pairwise top-8 overlap between subjects falls from 0.51 to 0.03; 119 of 190 subject pairs share no expert (Figure 1). Expert 224 wins one subject (politi-

cal science, by a 5.6% margin). The 122B generation block, exploratory because its trimmer failed, shows 18 distinct winners. The token-balanced control holds: with 446 prompt tokens and 2,048 generated tokens per packed prompt, the top-8 overlap between the prefill and generation profiles is 0.00, 0.23 and 0.07 across the three prompts by routed weight (0.07, 0.20 and 0.11 at top-15; 0.18, 0.28 and 0.22 at top-30), and per-token routing entropy is unchanged (0.958 versus 0.953) (Figure 2). The dispersion is a property of the regime; matching the token counts leaves it intact.

The official instruction-tuned checkpoint reproduces the shape. On its 100 prompts across eight types, prefill has three distinct winners (224 in six of eight; Herfindahl 0.59) and generation seven (Herfindahl 0.16); the pooled prefill and generation top-8 sets share no expert. Between-type overlap is 0.42 in prefill and 0.09 in generation.

### 3.3 The same subject recruits the same expert set

Inside a packed answer that walks through 20 subjects, the top-8 set changes at almost every subject boundary (Figure 3). Overlap between the two halves of one subject’s segment averages 0.49 ( $n = 46$  segments; the within-segment floor); overlap between different subjects in the same answer averages 0.053, 0.072 and 0.046 in prompts A, B and C (adjacent subjects alone: 0.047). The sets also return: archaeology recruits experts 191, 80 and 135 in A, in C and in the independent probe; neuroscience 54 and 24 in A, B and P; computer science 206, 207 and 189 in A, C and P; chemistry 130 leads in A, B and P; mathematics 100 in A, B and P.

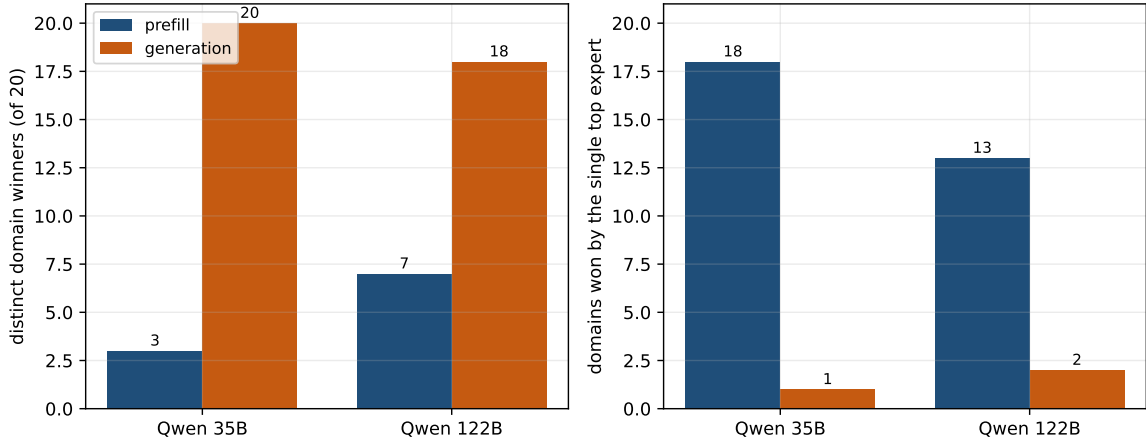
Table 2 gives the identification test. A subject’s generation-time set from the packed explanations (A) identifies that subject among the 20 independently asked subjects in 23 of 24 queries (0.96; Wilson 95% CI 0.80 to 0.99); from the packed synthesis (C), 36 of 38 (0.95; 0.83 to 0.99). Over all 168 bidirectional queries across the four designs, 117 are correct (0.70; 0.62 to 0.76); excluding the biography prompt, 73 of 86 (0.85; 0.76 to 0.91). The matched-subject Jaccard averages 0.39 (A to P) and 0.38 (C to P) against 0.03 for mismatched subjects; two random top-8 sets overlap at 0.017. Shuffling the subject labels 200 times gives a mean identification of 0.062 (95th percentile 0.095, maximum 0.119).

### 3.4 Controls

**Wording.** The identical procedure on the prefill rows of the same questions identifies the subject in 7 of 40 queries (A to P, 0.17; CI 0.09 to 0.32) and 6 of 40 (C to P, 0.15; 0.07 to 0.29); over all six design pairs, 73 of 240

| Checkpoint                        | Gating                  | Prompt set                                | Leading expert                                    | Wins         | Distinct | Herfindahl |
|-----------------------------------|-------------------------|-------------------------------------------|---------------------------------------------------|--------------|----------|------------|
| Qwen3.5-35B-A3B, HauhouCS         | softmax, 8 of 256       | 60 (20 subjects $\times$ 3)               | 224                                               | 18 of 20     | 3        | 0.815      |
| Qwen3.5-35B-A3B, official         | softmax, 8 of 256       | 100 (50 pairs, 8 types)                   | 224                                               | 6 of 8 types | 3        | 0.59       |
| Qwen3.5-35B-A3B, official         | softmax, 8 of 256       | 12 (5 subjects $\times$ 2 registers)      | 95 (224 next)                                     | 5 + 3 of 12  | 5        | 0.278      |
| Qwen3.5-122B-A10B, HauhouCS       | softmax, 8 of 256       | 60 (20 subjects $\times$ 3), routing-only | 233                                               | 13 of 20     | 7        | 0.445      |
| GLM-4.7-Flash, official           | sigmoid, 4 of 64 + bias | 300 (20 subjects $\times$ 15)             | 10                                                | 12 of 20     | 5        | 0.41       |
| GLM-4.7-Flash, HauhouCS (control) | sigmoid, 4 of 64 + bias | register battery, 140 cells               | set {10, 20, 28, 51} same set on both checkpoints | –            | –        | –          |

**Table 1:** Prefill replication, one row per checkpoint. Pooled over the prompt tokens, one expert leads prefill routing for most subjects on every checkpoint. Wins are subjects (or prompt types) taken by the leading expert; Distinct is the number of different winners in the list; Herfindahl is  $\sum_i p_i^2$  over the fraction of subjects won per expert. On the official 35B domain/register design, expert 95 wins 5 of the 12 prompts and expert 224 wins 3. The GLM HauhouCS row is a control with no subject axis: its pooled prefill top-4 set on the main register battery is {10, 20, 28, 51} on both the official checkpoint and the fine-tune (overlap 1.0), and {16, 20, 28, 51} on the two secondary batteries, again identical across checkpoints; expert 10, the winner on the subject battery, sits at rank 2 on the register battery. Rows 1 and 4 are the v1.1 Table 1 prefill rows.



**Figure 1:** Left: number of distinct experts among the 20 domain winners. Right: number of domains won by the single most frequent winner. In both models the prefill winners collapse onto a generalist and the generation winners spread across the domains.  $n = 60$  prompts per model (20 domains  $\times$  3). The 122B generation values are exploratory: that capture’s trimmer failed and about 45% of its generation block is post-answer spill (Methods, Provenance and scope).

(0.30). The one prefill pair that identifies well is A to C, 27 of 40 (0.68), where the two packed prompts share their carrier text, template and subject order. Prefill sets follow the question’s tokens; generation sets follow the subject of the answer. Span length does not explain the gap: using only the first 16 generated tokens of each packed segment, shorter than the question that produced it, identification is 22 of 24 (A to P) and 31 of 38 (C to P); with the first 64 tokens it is 23 of 24 and 35 of 38, the same as the full segments.

**Position.** Comparing A with C, whose subjects the model answered in a different order (the prompts list them identically; C regrouped its answer under thematic headers): different subjects at the same position overlap at 0.020; the same subject at different positions overlaps at 0.194. The subject wins 9 of 10 comparisons, with one tie at zero (history).

**Drift.** Restricting candidates to the previous, same and next subject in the target’s order, identification is 139 of 168 (0.83; CI 0.76 to 0.88) against an exact chance of 0.354; excluding B, 78 of 86 (0.91). Adja-

cent subjects overlap at 0.047. The sets change at boundaries rather than sliding along the answer.

**Register.** The biography prompt (B) holds the same subject labels but changes the requested task and voice. Pairs involving B identify at 44 of 82 (0.54; CI 0.43 to 0.64) against 0.85 for the other pairs. The subject expert persists (mathematics 100, neuroscience 54 and 24, chemistry 130) while the rest of the set follows the register. In the HVAC control, one fixed paragraph asked in a technical voice and in a first-person experiential voice gives a prefill top-15 overlap of 0.50 and a generation overlap of 0.00 ( $n = 6$  per register). Dialect is close to a no-op: AAVE and academic-English versions of matched prompts overlap at 0.64 to 0.71 in generation, above the 0.28 floor for unrelated prompts of the same type.

**Checkpoint.** On the official instruction-tuned 35B, in the domain/register design, generation sets pair prompts by subject in 11 of 12 nearest-neighbour queries (0.92; CI 0.65 to 0.99; chance 0.15, since finance contributes four prompts) with within-subject

| Pair (X→Y)                                | Shared subjects | Generation |                          |              | Prefill      |                          |              |
|-------------------------------------------|-----------------|------------|--------------------------|--------------|--------------|--------------------------|--------------|
|                                           |                 | Hits       | Accuracy [95% Wilson CI] |              | Hits (of 20) | Accuracy [95% Wilson CI] |              |
| A→B                                       | 12              | 7          | 0.58                     | [0.32, 0.81] | 7            | 0.35                     | [0.18, 0.57] |
| A→C                                       | 12              | 7          | 0.58                     | [0.32, 0.81] | 14           | 0.70                     | [0.48, 0.85] |
| A→P                                       | 12              | 11         | 0.92                     | [0.65, 0.99] | 4            | 0.20                     | [0.08, 0.42] |
| B→A                                       | 12              | 6          | 0.50                     | [0.25, 0.75] | 6            | 0.30                     | [0.15, 0.52] |
| B→C                                       | 14              | 5          | 0.36                     | [0.16, 0.61] | 6            | 0.30                     | [0.15, 0.52] |
| B→P                                       | 15              | 9          | 0.60                     | [0.36, 0.80] | 3            | 0.15                     | [0.05, 0.36] |
| C→A                                       | 12              | 7          | 0.58                     | [0.32, 0.81] | 13           | 0.65                     | [0.43, 0.82] |
| C→B                                       | 14              | 6          | 0.43                     | [0.21, 0.67] | 10           | 0.50                     | [0.30, 0.70] |
| C→P                                       | 19              | 18         | 0.95                     | [0.75, 0.99] | 4            | 0.20                     | [0.08, 0.42] |
| P→A                                       | 12              | 12         | 1.00                     | [0.76, 1.00] | 3            | 0.15                     | [0.05, 0.36] |
| P→B                                       | 15              | 11         | 0.73                     | [0.48, 0.89] | 1            | 0.05                     | [0.01, 0.24] |
| P→C                                       | 19              | 18         | 0.95                     | [0.75, 0.99] | 2            | 0.10                     | [0.03, 0.30] |
| A↔P                                       | 24              | 23         | 0.96                     | [0.80, 0.99] | 7 of 40      | 0.17                     | [0.09, 0.32] |
| C↔P                                       | 38              | 36         | 0.95                     | [0.83, 0.99] | 6 of 40      | 0.15                     | [0.07, 0.29] |
| All pairs excluding B                     | 86              | 73         | 0.85                     | [0.76, 0.91] | 40 of 120    | 0.33                     | [0.26, 0.42] |
| Pairs involving B                         | 82              | 44         | 0.54                     | [0.43, 0.64] | 33 of 120    | 0.28                     | [0.20, 0.36] |
| All pairs                                 | 168             | 117        | 0.70                     | [0.62, 0.76] | 73 of 240    | 0.30                     | [0.25, 0.37] |
| Shuffled-label null (mean; 95th pct; max) |                 |            | 0.062; 0.095; 0.119      |              |              | 0.050; 0.067; –          |              |

**Table 2:** Cross-design subject identification, per ordered design pair and collapsed. Designs: A (packed explanations, 12 segments), B (packed biographies, 15), C (packed synthesis, 19), P (independent probe, 20 subjects). A query segment in design X is assigned the subject of its highest-Jaccard candidate in design Y; ties count as misses. Generation denominators are the shared-subject counts (the twelve ordered pairs sum to 168); prefill rows contain all 20 questions in every design, so each ordered pair has 20 queries and each unordered pair 40. Law is absent from C; A and B are capped at 12 and 15 segments by the generation budget. Confidence intervals are two-sided 95% Wilson intervals. The shuffled-label null permutes subject labels across candidates 200 times (generation, NumPy `default_rng(11)`) and 100 times (prefill, seed 7). The two collapsed prefill rows (excluding B, 40 of 120; involving B, 33 of 120) are derived by addition from the per-pair counts.

overlap 0.46 against between-subject 0.035; prefill pairs 4 of 12 (0.52 against 0.47). Across the 50-pair run, a subject’s generation set on the official checkpoint identifies the same prompt type on the HauhouCS checkpoint in 7 of 8 types in both dialects, and the same legal (122, 109, 85), medical (206, 247) and identity (103, 139) experts lead on both checkpoints and both dialects. On identical prompts the fine-tune moves the generation top-8 set about as much as switching dialect on the same content does (overlap 0.70 between checkpoints against 0.75 between the two dialect versions of one prompt) and far less than switching prompt (0.08 across all prompt pairs); the movement concentrates in layers 0 to 2.

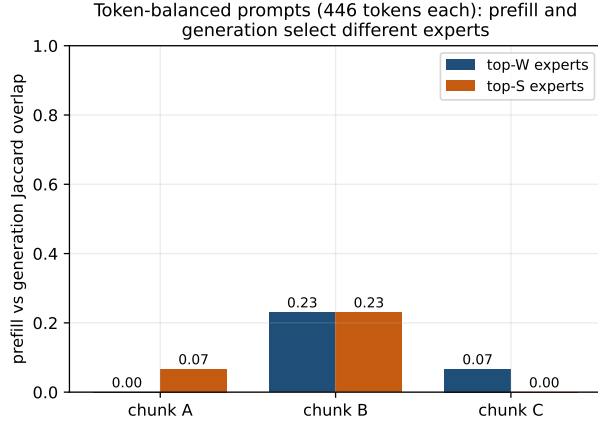
### 3.5 Scope

On a long, coherent technical paragraph followed by a technical answer (30 prompts, official checkpoint), prefill and generation share the set: overlap 0.60 to 0.78, the same leaders (166, 151, 41) on both sides, and per-token entropy 0.956 against 0.958. A narrow-topic prompt (medical triage) already routes to its topic set in prefill (247, 47; overlap with generation 0.23 to 0.37). The prefill/generation separation is strongest when a short question in one voice initiates a long answer organized around its subject.

## 4 Discussion

The two routing regimes reconcile the prior findings without rejecting any of them. The Standing Committee (Wang et al., 2026b), the fine-grained task experts of Herbst et al. (Herbst et al., 2026), the polysemantic single experts of Ye et al. (Ye et al., 2026), and the question-token-scored experts of Do et al. (Do et al., 2026) are all measurements on tokens supplied as input; our prefill numbers agree with each of them, on three checkpoints none of them tested (both Qwen3.5-35B-A3B checkpoints and the 122B) and on the GLM-4.7-Flash fine-tune under a second gating mechanism. Wang, Hayou and Nalisnick (Wang et al., 2026a) showed that prompt-level routing does not predict rollout routing and left the organization of the divergence open; the identification test answers it: for short mixed prompts, the rollout is organized by the local subject of the answer. Their two regimes are pairs of prompts that converge or diverge over a rollout; ours are the prompt and the answer of a single prompt. Their observation that which pair-regime applies “is not predictable from the prompt alone” is what a subject-organized rollout looks like from the pair side: pairs that share a subject stay similar, pairs that split subjects diverge.

The unit matters. Herbst et al. probe single experts and find operations; Ye et al. follow single tokens



**Figure 2:** Token-balanced control (35B model, three packed prompts of 446 prompt tokens and 2,048 generated tokens each). Jaccard overlap between the prefill and generation top-eight expert sets, ranked by routed weight and by selection rate. With token counts matched, the two regimes select largely disjoint experts.

across layers and find paths; we pool routed weight over a subject’s span and find sets that recur. These are three units, and the disagreement between “no domain specialists” and “domain specialists” is largely a disagreement about which one is being counted. A subject’s set may well be a coalition of fine-grained operation experts that co-occur when the model writes about chemistry; the causal test supports that reading. Biasing the four philosophy leaders off leaves 15 of 20 subject winners in place, per-subject overlap with baseline at 0.79, and the philosophy answers on topic and of the same length. The leaders are removable, so the paper’s claim is about routing organization, not about experts as the seat of subject knowledge.

Register is the second organizer. Holding subject fixed and changing the voice of the answer moves the generation set as much as changing the subject does, while dialect barely moves it. The 60-prompt probe holds register roughly constant, which is why its winners read as subject; the biography prompt is where register was allowed to move, and its identification rate is the measure of how much of the set register owns. A probe that crossed the two axes on the full subject list would separate them on one design rather than across three.

The mechanism proposed by Wang, Hayou and Nalnick, that routing is a linear readout of hidden-state geometry, is consistent with everything here and offers the simplest account of it: generated tokens about chemistry have chemistry-shaped hidden states, and the router reads them; prompt tokens in one voice have voice-shaped hidden states, and the router reads those. Their attribution of prefill collapse to reasoning preambles does not carry over: the flat default appears on a

non-reasoning fine-tune and on the official checkpoint with no chain of thought. Their rollout curves pool the prompt with the rollout; ours exclude it, which is why their divergence appears gradual while ours is present in the first 64 generated tokens. The title, then, disputes the scope of the myth claim rather than its mechanism: we reproduce the myth papers’ empirical result where they measured it, and the subject-organized generation we report is what their geometry predicts once the tokens being routed carry the subject.

## 5 Limitations

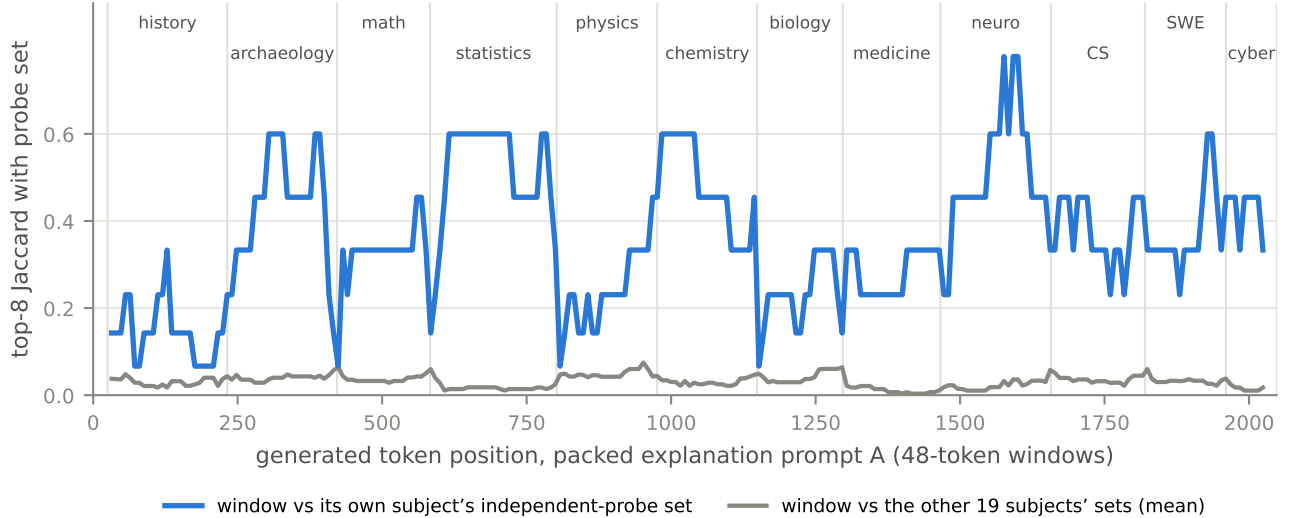
Three prompts per subject make the winner tables the weakest layer of the evidence. Only four generation winners on the 35B lead by a margin that survives dropping one prompt (philosophy, expert 114, 49% ahead; psychology, 146, 37%; mathematics, 100, 23%; linguistics, 103, 13%), and a worst-case bound allows 16 of 20 to change. The dispersion (Herfindahl 0.815 against 0.05) is far outside this noise; the identity of most named experts is not. Per-prompt tensors for the probe did not survive, so a leave-one-out is not possible; a 10-prompt-per-subject capture on the official checkpoint would close this.

The primary packed-generation evidence is one model, the HauhuCS Qwen3.5-35B-A3B. The official instruction-tuned checkpoint replicates the shape and the subject pairing on smaller designs (100 prompts across eight types; 12 prompts across five subjects). The 122B contributes a clean prefill replication and an exploratory generation result whose block is about 45% post-answer spill; its layer-pooled tables also mix the DeltaNet-mixing and standard-attention routing systems, whose top-8 expert vocabularies are disjoint on that checkpoint. GLM contributes the prefill half only; its generation captures have no subject axis. The cross-family claim is therefore for prefill, and the generation claim is for one architecture on two checkpoints.

Our “official” checkpoint is the instruction-tuned release; the pretrained base was not captured. The checkpoint comparison shows the organization persists across the uncensoring fine-tune, not that it precedes instruction tuning. The fine-tune moves routing behaviour by less than a prompt change does and leaves the GLM routers bit-identical; the Qwen router matrices were not compared directly.

Expert sets are layer-pooled by expert index, and expert 114 at one layer is a different module from expert 114 at another. Per-layer profiles on the probe show the generation dispersion at every layer, but per-layer identification on the packed designs is not available. Sets are defined by routed weight, which measures selection rather than what the selected expert writes; a





**Figure 3:** Within-answer top-8 overlap along the packed explanation prompt with subject boundaries marked. Each 48-token window’s top-8 set is compared with its own subject’s independent-probe set (mean 0.36) and with the other 19 subjects’ sets (mean 0.03); the own-subject overlap is higher in every one of 250 windows, and the dips fall at the boundaries.

residual-write-norm variant is a cheap follow-up. Prefill spans for the packed prompts were rescaled across a tokenizer mismatch (391 to 392 versus 446 tokens) and may be off by one or two positions; generation spans are exact.

Finally, the regime effect is scoped: on a long coherent technical prompt the prefill and generation sets coincide. The claim is for short mixed prompts that initiate longer subject-organized answers.

## 6 Conclusion

Where you measure routing determines what you see. Measured on the tokens a model reads, expert routing converges on a small default set whatever the subject, and we reproduce that result on five checkpoints across two families and two gating mechanisms. Measured on the tokens the same model writes, the same routers organize by the local subject of the answer: sets stay put within a subject, turn over at its boundary, and return for the same subject under a different prompt design at 95 to 96% identification against 6% by chance on the fine-tuned checkpoint, and pair prompts by subject in 11 of 12 cases on the official one. Prior work measured the reading. Specialization is visible in the writing.

## Data and Code Availability

The data supporting this paper are in the curated repository `expert-specialization-data` (commit `f06cd3e`), whose `PROVENANCE.md` gives the run id, model file, capture binary, decode flags and source path for every set, and whose `MANIFEST.sha256` covers every file. That repository’s provenance records, manifests, build script, and complete analysis scripts are included with this deposit under `curated-data-index/`; the full archive (about 6 GB of router tensors and per-cell compactions) is available from the author. The version 1.1 supporting data under `data/` (the 60-prompt primary probe, the token-balanced control, and the 122B replication) remains included. The 35B primary probe retains per-subject, per-layer  $W/S/Q$  summaries; the packed prompts retain per-token, layer-averaged profiles and generated tokens; the official-checkpoint, HVAC, register, dialect and GLM captures retain raw per-layer router tensors or per-cell compactions with the sha256 of each original. The identification test, the four nulls, the per-pair table, the GLM control sets and the checkpoint-shift analysis are the scripts in `audit_20260828/`; the 2026-08-28 audit findings, the frozen predictions written before it, the hostile review and the number audit of this draft are in the paper repository’s `ledger/`. No script re-runs a model.

# AI Collaboration Statement

Generative AI (Anthropic’s Claude) was used as a tool in preparing this manuscript: to assist with drafting and revising prose, including first drafts of the Results, Discussion, Limitations, and Conclusion sections written to the author’s analyses and outlines; to audit every reported number against its source capture; and to organize the manuscript and locate and format the bibliography. All experiments, captures, and numerical results are the author’s; every reported value was traced to a source capture (see `SOURCES.md` and the `ledger/` audit trail), and every reference was verified against its authoritative record. No AI system is an author. The author reviewed the entire text and takes full responsibility for its content.

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This work received no external funding. The author declares no competing interests. The models studied are public checkpoints, official releases and third-party fine-tunes, used as-is for measurement; the author has no affiliation with their creators.

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## A Per-layer winner counts

Distinct subject winners per layer on the independent probe (HauhuCS Qwen3.5-35B-A3B, 20 subjects, three answers each; winner = the expert with the highest layer-specific  $W$  for the subject). Generation: mean 14.2 distinct winners per layer (range 8 to 18); prefill: mean 1.6 (range 1 to 6).

| Layer | Prefill | Generation | Layer | Prefill | Generation |
|-------|---------|------------|-------|---------|------------|
| 0     | 1       | 14         | 20    | 1       | 18         |
| 1     | 1       | 8          | 21    | 1       | 16         |
| 2     | 2       | 9          | 22    | 1       | 13         |
| 3     | 1       | 8          | 23    | 1       | 11         |
| 4     | 3       | 16         | 24    | 2       | 14         |
| 5     | 1       | 16         | 25    | 1       | 17         |
| 6     | 1       | 14         | 26    | 1       | 13         |
| 7     | 1       | 16         | 27    | 3       | 14         |
| 8     | 1       | 18         | 28    | 1       | 15         |
| 9     | 1       | 15         | 29    | 1       | 13         |
| 10    | 1       | 14         | 30    | 2       | 16         |
| 11    | 1       | 13         | 31    | 2       | 17         |
| 12    | 3       | 14         | 32    | 1       | 12         |
| 13    | 2       | 16         | 33    | 1       | 11         |
| 14    | 1       | 14         | 34    | 1       | 13         |
| 15    | 5       | 15         | 35    | 3       | 18         |
| 16    | 1       | 10         | 36    | 1       | 16         |
| 17    | 2       | 15         | 37    | 1       | 17         |
| 18    | 2       | 14         | 38    | 1       | 16         |
| 19    | 1       | 17         | 39    | 6       | 11         |

**Table 3:** Distinct subject winners per layer, prefill versus generation, independent probe.